Synthetic Relativity: Tri-modality and the Fixpoint Semantics of the Logical Observer

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Introduction to Synthetic Relativity

Synthetic Relativity

Scope of talk

- ‘Synthet axiomaticic’ as in synthetic geometry, or synthetic domain theory, i.e. *axiomatic*
- **Focus:** Self-reflection and reasoning about theories of action

Out of scope

This is *not* a first-order logic foundation of physical relativity theories with accelerated observers e.g. work by the *Budapest Relativity Research Group* (Madaráasz et al. 2006, Székely et al. 2010)

- André et al., 2013 provides an axiomatisation complete for Lorentzian manifolds, i.e. the Lorentz interval invariant $(c\Delta t)^2 - distance^2$ always equals zero.

Self-reflective reasoning about action

Motivation: Understand role of self-reflection for reasoning about action, for tasks such as

- Classical planning,
- Generalised planning,
- Strategic reasoning (strategy synthesis, winnability, etc.).

Reach goal:

- Self-transcending reasoning, i.e. autonomous theory progression for transcending a theory of action toward solving *new problems*

Transformation example in logic

The space-time tradeoff **Memoisation** versus **lazy evaluation**.

Memoisation (Time $\longrightarrow$ Space)

You take a proof that requires coinductive steps (Time), evaluate it in advance, and cache the valid answers into a lookup table (Space/Induction)—You have contracted logical time (proof) and dilated logical space.

Lazy Evaluation (Space $\longrightarrow$ Time)

You take memory (Space/Induction) and collapse it. Instead of storing the data, you store a coalgebric rule that generates proofs sequentially, only when requested (Time/Coinduction)—You have contracted logical space and dilated logical time (proof).

The logical observer

Different logical observers evaluate systems differently:

- **One observer** (evaluating lazily/coinductively) stretches out the "time" of the proof.
- **Another observer** (evaluating eagerly/inductively) compresses the proof into into memory "space".

However, the fundamental computational truth—the realisation—cannot change.

If truth were not invariant under these frame-shifts, logic would collapse into inconsistency. The spatial distortion of the data structure and the temporal delay in evaluation mathematically cancel out to preserve the invariant **realisation**.

Logical interval invariant?

In synthetic relativity, inductive structure and coinductive process are two mathematically intertwined ways of evaluating the same invariant truth.

- **Induction** (static space/structure) and
- **Coinduction** (unfolding time/process) are the coordinate axes, and
- **Realisation** is the invariant interval.

Rotating logical frame of reference changes how much proof exists in spatial topology versus how much unfolds dynamically in computational time, but invariant interval—the **realisation**—is preserved.

Note in the **situation calculus** that since *regression* is trivially **confluent**, as per Church-Rosser theorem as it is deterministic, the order in which you evaluate it does not change its final semantic truth, so the **realisation** is preserved.

Inference and Categorical Duality

Inference duality of the observer

Logical Benders decomposition (Hooker & Ottosson, 2003) shows how the observer iteratively proposes a model of itself and verifies it via self-reflection.

(Davies, Pearce, Stuckey & Lipovetzky 2015) showed this to be competitive with state of the art planners on International Planning Competition (IPC) benchmarks

- **The Master (The Proposer)**: The observer proposes a relaxed model of its current state or plan. Operator (action) counting using MIP
- **The Subproblem (The Verifier)**: The observer reflects on this proposal, testing its logical consistency with respect to the overall problem. Sequencing actions using SAT
- **The Cut (The Inference)**: When the proposal fails, the observer generates a logical inference (a Benders cut) that restricts its future reasoning. Conflict driven clause learning (SAT).

Proven to converge to the optimal feasible solution. It can be modelled as finding a fixpoint of a monotonic constraint-generation operator

Inference duality of the observer - Limitations

It does not compute a “least fixpoint” in the traditional sense as it intentionally stops early once optimal feasibility is proven (it is based on propositional logic)

- Note that the **situation calculus** *does* rely on least fixpoint in first order logic.

Property Persistence as a Greatest Fixpoint

The Challenge: Verifying universally quantified temporal queries (e.g., "ϕ persists indefinitely") requires second-order induction, which strictly limits automated first-order reasoning.

The Single-Step Operator: (Kelly & Pearce, AIJ 2010) captures the condition for ϕ holding *now* and being preserved after *any single legal action* via regression ($\mathcal{R}$):

$$\Gamma(Z) \equiv \phi(s) \land \forall a \big(Poss(a, s) \to \mathcal{R}[\, Z(do(a, s)) \,] \big)$$

Greatest Fixpoint (νZ):

- Persistence is an infinite-horizon **safety property**.
- The persistence condition is the **greatest fixpoint** of this operator: $\nu Z.\ \Gamma(Z)$.

Result: Iteratively applying this meta-level calculation computes the maximal invariant without explicitly invoking the induction axiom.

- By wrapping the single-step regression operator inside a monotonic formula transformer, we evaluate the future back in the current state.
- Taking the greatest fixpoint of this regression operator yields the exact, most permissive preconditions required for the property to persist indefinitely using only first-order logic.

Strategy Synthesis in Multi-Agent Games

- **Scaling Fixpoint Framework:** Property persistence scales natively from single-agent persistence to adversarial settings via *Situation Calculus Synchronous Game Structures (SCSGSs)*.
- **General Game Playing (GDL):** (De Giacomo, Lespérance & Pearce, ECAI 2016) model joint, synchronous multi-player moves over infinite domains via a single `tick(m_1, ..., m_n)` action.
- **First-Order Alternating-Time μ-Calculus:**
  - **Safety:** Greatest fixpoints (νZ) — Generalising persistence to maintain a winning condition against any opponent (*Weak Winnability*).
  - **Liveness:** Least fixpoints (μZ) — Enforcing reachability to force a win in a finite number of steps (*Strong Winnability*).
- **Synthesis:** By upgrading the quantifier prefix to alternating game quantifiers ($\exists \vec{m}_{agt} \forall \vec{m}_{env}$), interleaving regression with fixpoint approximations automates strategy synthesis.

The exact same regression-based fixpoint mathematics we used for property persistence natively empowers the μ-calculus. - By upgrading the quantifier prefix from ‘nature does anything’ to alternating game quantifiers, we can use both greatest and least fixpoints to automatically synthesise strategies and guarantee winnability in infinite-state games.

Categorical duality

Is property persistence just an algorithmic trick with regression?

No. It represents a fundamental categorical duality in action theories.

- **The Inductive Action Theory (Standard SitCalc):**
  - Builds state histories forward from S_0.
  - Categorically maps to **Initial Algebras** and **Least Fixpoints** (μ).
  - Naturally solves *reachability* and *liveness*.
- **The Coinductive Persistence Theory (AIJ 2010):**
  - Persistence is an infinite-horizon safety property.
  - Categorically maps to **Terminal Coalgebras** and **Greatest Fixpoints** (ν).
  - Employs **Regression** (the *right adjoint* to progression) to compute maximal invariants.

Limitations

What are the limitations of our approach?

- Regression is handled *separately* by different logics: μ calculus (greatest fixpoint μ and least fixpoint ν) $+$ situation calculus (least fixpoint μ)
- Neither the μ calculus nor the situation calculus captures *inference duality* or *categorical duality* explicitly
- They rely on a hand-coded basic action theory (BAT)

Recall **Stretch goal:** self-transcending reasoning

- A relativistic framework for reasoning about limits and topological structure
- A framework for the autonomous progression of action theories
- A partial order approach to both synthesising and hypothesising

Synthetic relativity axioms

Tri-modality & ternary representation

Because the categorical dual of a reflective system cannot be handled in a strict boolean space without losing information, a ternary representation becomes geometrically and logically necessary.

Standard provability logics like GLB (Gödel-Löb Bimodal Provability Logic) rely on absolute binary provability: xRy assumes an absolute background space where y is provable from x.

- In a ternary structure (drawing from Routley-Meyer semantics for substructural, relevance, and intuitionistic logics), you have a relation $R(x, y, z)$.
- This allows us to decouple the synthetic transformation from the topological space. The ternary relation natively introduces the Observer's frame. It reads: “Relative to reference frame x, the accumulation of inductive evidence y yields the coinductively realised semantic limit z.”

Synthetic Relativity: The Structural Axioms

Dynamic Trimodal Logic with Ternary Relational Semantics: The framework replaces absolute global topologies with a relativistic reasoning architecture governed by foundational structural postulates.

Axiom 1: The Ternary Relational Axiom Logical entailment is governed by three interacting modalities relative to a specific baseline Reference Frame (S_f):

- RDo **(The Subproblem / Syntax): The Inductive Accumulator ($\Box_1$)** Operates in constructive, non-Boolean logic. Recursively gathers finite, computationally observable properties (compact elements) along a directed logical path. RDo is equivalent to the $\Box$ modality of bimodal provability logic GLB: *“There exists a finite, step-by-step mechanical proof of this property.”*

Constructive Accumulation & Realisation

Axiom 2: The Axiom of Constructive Accumulation Within a given reference frame (S_f), the Relativistic Engine (PDo) iteratively drives the state forward, nesting inductive syntax (RDo) and semantic coinductions (ODo):

$$\mathrm{PDo}(S_r, S_o, S_f)$$

$$\mathrm{PDo}\Big(\,\mathrm{RDo}(R_1, S_r),\;\; \mathrm{ODo}(O_1, S_o),\;\; S_f\,\Big)$$

$$\mathrm{PDo}\bigg(\,\mathrm{RDo}\Big(R_2, \mathrm{RDo}(R_1, S_r)\Big),\;\; \mathrm{ODo}\Big(O_2, \mathrm{ODo}(O_1, S_o)\Big),\;\; S_f\,\bigg)$$

$$etc.$$

Theorem I: Semantic Realisation

The Synthetic Transformation

Axiom 3: The Transitivity Axiom Once Realisation is achieved, the relativistic reference frame shifts. The categorically preserved invariant (the semantic limit, O) becomes the new reference frame (S_f) for the next order of progression.

$$\mathrm{PDo}(N, O, S_f) \longrightarrow \mathrm{PDo}(X, Y, \mathbf{O})$$

- **Left Side:** The fully realised state within the initial reference frame (S_f).
- **The Arrow ($\longrightarrow$):** The Synthetic Transformation (Topological closure and Change of Base).
- **Right Side:** A new, higher-order autonomous progression begins. The complex limit O has mathematically synthesised into an atomic, discrete baseline coordinate for the new subproblem (X)

The Dual Fixpoints

Unifying Reachability (μ) and Persistence (ν)

By natively intertwining syntax (RDo) and semantics (ODo), the Relativistic Engine mathematically fuses two distinct forms of logical progression:

1. **Induction / Least Fixpoint (μ):** Governed by the RDo modality. The bottom-up synthesis of finite evidence from baseline ignorance ($\perp$). It guarantees computational *Reachability*.
2. **Coinduction / Greatest Fixpoint (ν):** Governed by the ODo modality. The top-down evaluation of continuous semantics. It guarantees *Safety and Property Persistence*.

Algebraic Compactness ($\mu = \nu$) Theorem I (Semantic Realisation) formally computes a **Bilimit**. At the exact coordinate of convergence, the active inductive trace (μ) perfectly fulfills the persistent semantic

Synthetic relativity logic

The Observer’s Dual Logic (Syntax vs. Semantics)

To execute our modalities, we require an architecture that natively fuses classical (inductive) logic with intuitionistic (coinductive) logic.

- **Memoisation (Induction / The Actor):** Constructive in the *data* sense. It explicitly builds and stores finite, historical knowledge. It captures logical invariants and acts as the proof of what has been physically done.
- **Lazy Evaluation (Coinduction / The Observer):** Constructive in the *intuitionistic* sense. A “thunk” (an unevaluated promise) is an observable property. The runtime intuitively maintains the topology of an infinite stream, only verifying when locally forced.
- **The Incompleteness Boundary:** Logic remains classically monotonic locally. However, due to relative incompleteness (Gödel’s Incompleteness & Beklemishev’s reflection principles), a

The Foundational Logic: Domain Theory in Logical Form

To map this topology of reasoning, we ground our system in Abramsky's **DTLF** (1991). We abandon the absolute "God's eye" Boolean view ($A \vee \neg A$) for an endogenous, inside-out construction of truth.

- **Baseline Ignorance ($\perp$):** Computation does not start at "False"; it starts at Bottom ($\perp$)—a mathematical coordinate of zero information.
- **Compact Elements:** Variables are not absolute binary states; they are discrete, finite pieces of verifiable evidence (e.g., *a specific sensor reading*).
- **Directed Accumulation:** Knowledge is actively gathered. Because you cannot empirically "observe" an infinite negative in finite time, DTLF relies on a strictly positive logic, generating an upward, directed proof path ($\sqsubseteq$).

Spatial Completeness & The Static Limitation

The profound mathematical power of vanilla DTLF lies in **Spatial Completeness** (rooted in Stone Duality).

- **The Guarantee:** It proves a 1-to-1 isomorphism between the algebraic, step-by-step **Syntax** ($\phi \vdash \psi$) and the continuous, geometric **Semantics** ($[\![\phi]\!] \subseteq [\![\psi]\!]$).
- **Catching Limits:** Computations resolve exactly when the directed path of finite evidence mathematically catches an infinite continuous topological limit (a Supremum).
- **The Flaw (The Crisis):** Vanilla DTLF perfectly describes a single, *static* universe. To untangle induction from coinduction and model dynamic, continuous learning, **the Reference Frame itself must dynamically shift.**

Encoding Planning in Synthetic Relativity (R-DTLF)

Classical planning (e.g., Situation Calculus) relies on a global, static Basic Action Theory (BAT), where rigid evaluation across absolute history leads to combinatorial explosion. In **R-DTLF**, planning is constructed dynamically:

- **Initial Database** $\rightarrow$ **Reference Frame (**S_f**):** Dictates the local inductive bias—bounding exactly which *fluents* (e.g., Disk positions) are currently observable, replacing the rigid global space.
- **The Situation Term (**$do(a, s)$**)** $\rightarrow$ **Inductive Accumulation (**RDo**):** Instead of an endlessly nested string of rigid state-jumps, actions are encoded as verifiable proof steps recursively accumulating along a directed path.
- **Goal Formula (**$\exists s.\, G(s)$**)** $\rightarrow$ **Semantic Target (**ODo**):** The goal is not a static Boolean theorem. It is encoded as a **Target Semantic**

Execution & Invariant Synthesis

Towers of Hanoi: Solving Combinatorial Explosion

In standard classical planners, the Towers of Hanoi puzzle has a continuous branching factor of 3, leading to intractable combinatorial explosion for deep goals.

1. Local Unrolling (Building the Situation Trace): The engine recursively accumulates primitive actions (moving disks), generating the topological equivalent of a nested $do()$ sequence:

$$\mathrm{PDo}\Big(\ \mathrm{RDo}\big(\mathrm{move}(D_1, P_2), \ldots S_f\big),\ \ \mathrm{ODo}\big(O_{\text{target}}, \mathrm{Goal}\big),\ \ S_f\ \Big)$$

2. Limit Catching (The Parity Invariant): Through top-down semantic reflection ($\Box_\omega$), the ODo modality mathematically coinduces a structural property of the optimal traces: **The Parity Invariant** (

O) *(The smallest disk D must cycle consistently, and alternate*

Concrete Syntax: Visualising the Encoding

Translating Continuous Syntax into Classical Axioms

How does the continuous logic of R-DTLF physically bypass the Frame Problem and compile heuristics?

1. Sidestepping the Frame Problem (SSA vs. R-DTLF)

- **SitCalc SSA (Global Tracking):** Evaluates the entire history (s) across rigid Successor State Axioms:

$$On(d, p, do(a, s)) \equiv a = \text{move}(d, p) \vee \big(On(d, p, s) \wedge \neg\exists p'.\, a = \text{move}(d,$$

- **R-DTLF (Local Entailment):** Operates purely on verifiable observables. Irreleva global variables are topologically invisible (no frame axioms are computed duri unrolling):

$$\text{RDo}\big(\text{move}(d, p), S_f\big) \vdash On(d, p)$$

Bridging Worlds: Endogenous Logic Compilation

R-DTLF does not discard classical logic; it functionally *generates* it at the semantic limits.

Theorem III: Topological Booleanization (Compiling Successor State Axioms)

Upon semantic realisation, the continuous intuitionistic trace generated by RDo can be losslessly compiled into a discrete, classical First-Order baseline via the **Lawvere-Tierney Double Negation ($\neg\neg$) Topology**. Topos theory formally guarantees that inside every fluid, intuitionistic space exists a rigid Boolean sub-lattice that can be safely projected onto.

- **Compiling Successor State Axioms (SSAs):** This geometric compression is mathematically homologous to Ray Reiter's famous **Explanation Closure**. The Relativistic Engine acts as an endogenous logic compiler—synthesising complex discovered invariants directly into rigid First-Order logical rules.
- **Hybrid Execution:** Because of this formal theorem, we can securely export newly learned R-DTLF heuristics directly into standard

Comparison

Synthetic Relativity versus Situation Calculus versus GLB

Aspect	*Synthetic Relativity (R-DTLF)*	*Situation Calculus (BATs/SSAs)*	*Bimodal Provability logic (GLB)*
Modalities	RDo, ODo & PDo		
Ordering	Sequences of finite Posets, non-unique predecessors, non-well ordered	Successor function, Unique predecessors, non-well ordered	Transfinite ordinality: limits, Non-unique predecessors, well-ordered
Compactness?	Yes (Algebraic compactness)	No	No
Completeness?	Both spatially complete (retains continuous geometry) and Turing complete (gains universal computation via recursive domains).	Turing complete (can compute anything), but neither spatially complete (blind to continuous geometry) nor arithmetically complete (fails Gödel's test).	Arithmetically complete (flawlessly models the bounds of provability).

Appendix

Domain Theory in Logical Form (DTLF)

Introduced by Samson Abramsky (1991), DTLF provides the foundational “physics” of computation. To map the true topology of reasoning, we must first abandon classical, absolute logic.

- **The Classical Flaw:** Standard logic (Boolean) assumes an absolute, “God’s eye” view of reality (True/False). It relies on the Law of Excluded Middle ($A \lor \neg A$) and ignores *how* a system physically computes truth.
- **Intuitionistic & Constructive:** DTLF operates from the inside out. Truth is not a static given; it must be actively constructed step-by-step.
- **The Logic of Partial Information:** In DTLF, variables do not represent absolute boolean states; they represent *observable properties* and evolving states of partial knowledge.

The Scott Domain

In DTLF, computation is an upward, continuous climb from baseline ignorance toward a fully realised concept within a geometric **Scott Domain**.

- **Baseline Ignorance ($\perp$):** The logic does not start at "False." It starts at Bottom ($\perp$)—a formal mathematical coordinate representing zero information and zero computation.
- **Compact Elements (The Primitives):** The finite, discrete pieces of verifiable evidence a system can physically observe (e.g., *a specific sensor reading*, or *a sequential token*).
- **Directed Accumulation:** Knowledge is actively gathered. The system continuously accumulates these compact elements, generating a directed proof path ($\sqsubseteq$).

Spatial Completeness

The profound mathematical power of vanilla DTLF lies in Abramsky's **Spatial Completeness Theorem**, rooted in Stone Duality.

It proves a mathematically perfect, 1-to-1 isomorphism between:

1. **The Syntax:** The algebraic, step-by-step logical code ($\phi \vdash \psi$).
2. **The Semantics:** The continuous, geometric space of concepts ($[\![\phi]\!] \subseteq [\![\psi]\!]$).

The Guarantee: Every step of logical deduction perfectly corresponds to navigating a continuous topology. Computations resolve exactly when the directed path of evidence catches a continuous topological limit (Supremum, $\bigsqcup$).

Limitation: Vanilla DTLF perfectly describes a single, static universe. To model continuous learning and dynamic abstraction, the Reference Frame

The Core Philosophy

Exogenous Truth vs. Endogenous Observation

- **Classical Logic is Exogenous:** It looks at abstract, eternal truths from the outside.
- **Vanilla DTLF is Endogenous:** It is built from the inside out.
- **Observable Properties:** In DTLF, a formula ϕ does not represent an absolute "Truth." It represents a finite, computationally *observable property* (a compact open set in the Scott Topology).
- **The Act of Verification:** To assert ϕ is to mathematically state: *"Based on a finite amount of inductive computational steps, I have actively gathered the evidence to verify property ϕ."*

The Formal Syntax

A Strictly Positive Propositional Logic

Because a system can only assert what it can finitely verify, the baseline syntax is strictly positive:

$$\phi, \psi ::= \mathbf{t} \mid \mathbf{f} \mid \phi \wedge \psi \mid \phi \vee \psi \mid \mathrm{Constructors}(\phi)$$

- **t (True / Baseline):** The empty observation. Requires zero information (topologically, the bottom $\perp$ of the domain, where you know nothing).
- **f (False / Contradiction):** Impossible information (a computational crash).
- $\phi \wedge \psi$ **(Conjunction):** The active accumulation of evidence (both verified).

The Engine & Catching Limits

Entailment ($\vdash$) and Syntactic Fixpoints

Instead of static Boolean truth tables, DTLF uses an **Entailment Relation** to mathematically move "up" the directed poset towards a limit. $\phi \vdash \psi$: *"Any state of information satisfying ϕ automatically contains enough information to satisfy ψ."* - **Rules:** Reflexivity ($\phi \vdash \phi$), Transitivity (if $\phi \vdash \psi$ and $\psi \vdash \chi$, then $\phi \vdash \chi$), Conjunction ($\phi \wedge \psi \vdash \phi$).

How Vanilla DTLF Catches Limits: Because syntax is finite, an infinite continuous limit cannot be a single finite formula. - **Limits as Filters:** A semantic limit is a *Filter* or *Ideal*—an infinite, directed set of formulas mutually entailing each other ($\phi_1 \vdash \phi_2 \vdash \phi_3 \ldots$). - **The Fixpoint Operator (μ):** To handle recursive loops, vanilla DTLF relies on a static, syntactic variable axiom ($\mu X.\, F(X)$) to declare that a least fixpoint exists.

Categorical (Stone) duality: Domain Theory in Logical Form (DTLF) & Scott topologies

Through Stone duality, DTLF proves that these two sides are mathematically isomorphic:

- The Spatial Side (Semantics): Domains of computational values equipped with the Scott topology.
- The Algebraic Side (Syntax): The logic of observable properties, structured as an algebraic lattice.

To capture the continuous reality of the logical observer, we must transition to the semantics of **DTLF**.

Abramsky, S. (1987). *Domain theory in logical form*. Proceedings of the Symposium on Logic in Computer Science (LICS ’87), 47-53. IEEE.

Abramsky, S. (1991). *Domain theory in logical form*. Annals of Pure and

DTLF spaces are T_0 (Kolmogorov) spaces (Why it is non Hausdorff)

For a topological space to be Hausdorff (T_2), you must be able to take any two distinct points x and y and put them in completely separate, non-overlapping “bubbles” (disjoint open sets).

In domain theory, points are not independent locations in space; they represent stages of computation ordered by information content ($x \sqsubseteq y$ means “x is a partial approximation of y”).

Because open sets in DTLF represent observable properties, the Scott topology dictates that every open set must be upward-closed. If an observable property is true of a partial computation x (e.g., “it has printed the letter A”), it must remain true for any further refinement y (e.g., “it has printed A, B, and C”). Information is monotonic; it cannot be “un-learned.”

Compactness & finite observability

The syntax (the logic): Bounded Distributive Lattices

The semantics (the geometry): Spectral spaces

Compact open sets & finite observability:

Compactness bridges infinite mathematical semantics with finite computational syntax.

Qualities of DTLF

The Scott Topology perfectly matches a formal, inductive logic: Open sets in the Scott Topology The Logic (Syntax): A formula in standard DTLF represents a finite, verifiable piece of information. The Space (Semantics): This formula maps perfectly to a compact open set (a finite approximation) in the Scott Topology of the domain. The Proof Paths: Limit creation in the domain (taking the supremum of a directed set) corresponds exactly to logical entailment.

- Nets indexed by directed sets.
- Compact elements

A compact element (sometimes called a finite element). An element c is compact if, whenever c is less than or equal to the limit of a directed set, c is actually less than or equal to some element already inside that directed set.

Sequences of Finite Posets (SFPs): Partial ordering

Sequences of Finite Posets (SFP), often called Bifinite domains, were introduced by Gordon Plotkin in 1976, they are domains constructed entirely as the limit of finite approximations. They are the ‘gold standard’ in DTLF because they are robust enough to model complex features like higher-order functions and non-determinism, while retaining the exact topological properties—compactness—required for finite observability and Stone Duality

- In Computation (Semantics): It models the growth of information ($x \sqsubseteq y$ means x approximates y).
- In Topology (Scott Topology): It dictates what is observable (properties must be “upward-closed” along the order).
- In Logic (Syntax): It defines proof rules and deduction ($A \leq B$ means A entails B).

Comparison with Inductive Logic Programming (ILP)

To prove A relative to B with respect to C is to reject the notion of an absolute, context-free proof. In this ternary relation, C (S_r) serves as the topological Reference Frame (the environment or inductive bias). A (RDo) is the active, syntactic accumulation of finite evidence. B (ODo) is the evaluated semantic limit.

Therefore, the proof formally guarantees that within the strict geometric bounds of frame C, the evidence A is structurally sufficient to synthesise the invariant meaning B.

- Unlike classical Inductive Logic Programming (ILP) where background knowledge is static and human-coded, or standard ML where inductive bias is fixed globally by architecture, the Synthetic Relativity framework is dynamic.
- The system's background knowledge (Frame C) continuously evolves via the Transitivity Axiom, allowing the network to

Comparison with Inductive Logic Programming (ILP) (continued)

Gordon Plotkin proved in 1971 that the Relative Least General Generalisation (RLGG) of two clauses—relative to a background theory of arbitrary Horn clauses—may not exist as a finite clause under θ-subsumption.

- Inverse entailment: grounding background knowledge relative to a specific example, IE constructs the most specific finite clause that entails the example. The search space for the algorithm is then strictly bounded between the empty clause (the most general) and this finite bottom clause (the most specific), completely circumventing the need to generate an infinite RLGG (Muggleton, 1995).
- Extracting structure from higher order ILP (Cropper et al. 2020))